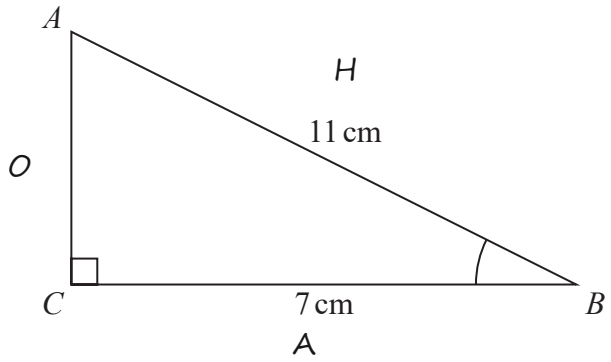
**5** $ABC$ is a right-angled triangle.




(a) Work out the size of angle $ABC$.
Give your answer correct to 1 decimal place.

Relative to angle B: adjacent = 7; hypotenuse = 11.

$\cos B = \frac{7}{11}$

**1 mark**
Use adjacent ÷ hypotenuse

$B = \cos^{-1}(7 \div 11) = 50.4788...°$

**1 mark**
50.5° to 1 decimal place

50.5 °

**(2)**

The length of the side $AB$ is reduced by 1 cm.

The length of the side $BC$ is still 7 cm.
Angle $ACB$ is still 90°

(b) Will the value of cos $ABC$ increase or decrease?
You must give a reason for your answer.

Increase: the new hypotenuse is 10 cm.

$\frac{7}{10} = 0.7$, which is greater than $\frac{7}{11} \approx 0.6364$.

**(1)**

**1 mark**
Compare the new and old cosine values

**(Total for Question 5 is 3 marks)**

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